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Gravity-wave momentum fluxes induced by multiple convective cells

Jaemyeong Mango Seo^{1*} and Jong-Jin Baik¹

Abstract

Convectively forced gravity waves (CGWs) play important roles in the atmosphere. CGW drag parameterizations for use in large-scale models typically represent subgrid-scale convection as a single convective source. This study uses an analytic framework to investigate how horizontal organization of multiple convective cells modifies the vertical flux of horizontal momentum. It is shown that the per-cell momentum flux induced by multiple convective cells is expressed as the momentum flux induced by a single convective cell times $[1 + J_n(\beta, \delta)]$. Here, $J_n(\beta, \delta)$ is the nonlinear modification factor arising from mutual interactions among distinct convective cells, where β is the ratio of compensating-cooling width to heating-core width, δ is the nondimensional spacing between the horizontal centers of convective cells, and n is the number of cells. Closely packed deep convective cells with broad cooling (large β and small δ) act to enhance the per-cell momentum flux relative to a single convective cell, whereas more widely spaced convective cells act to reduce it. As the number of convective cells increases, the region of positive J_n in the parameter space of β and δ shrinks toward smaller δ . These results indicate that purely geometric properties of convective organization (β , δ , and n) systematically modify CGW momentum flux. The proposed nonlinear modification factor can be easily incorporated into existing CGW drag parameterizations.

Keywords Stably stratified atmosphere, Multiple convective cells, Convectively forced gravity waves, Momentum flux, Gravity-wave drag parameterization

Introduction

Convectively forced gravity waves (CGWs) are known to play important roles in the atmosphere. For example, CGWs play a central role in maintaining the circulation in the stratosphere and mesosphere through their momentum transport and drag (Fritts and Alexander 2003). Because general circulation models cannot fully resolve the spectrum of gravity waves, parameterizations of subgrid-scale convective sources are required to represent gravity-wave effects (Kim et al. 2003). These

parameterizations are essential to realistically simulate large-scale phenomena, such as the quasi-biennial oscillation (QBO) (Dunkerton 1997; Schirber et al. 2015) and the polar night jet (Choi and Chun 2013). However, the differences between models and observations in tropical momentum flux and the sensitivity of the QBO to convective source specification are still large (Liu et al. 2014; Garfinkel et al. 2021).

CGW drag (CGWD) parameterizations typically treat subgrid-scale convection as a single representative source per grid cell. The approaches for representing CGWD include the parameterization which relates convective forcing to gravity-wave momentum flux through analytic formulations (Chun and Baik 1998, 2002; Song and Chun 2005; Choi and Chun 2011) and the parameterization which specifies a source spectrum from convective

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characteristics (Beres et al. 2004). These parameterizations calculate the cloud-top momentum flux. Comparisons against observations for the QBO show mixed results: some of the QBO structure are captured, but discrepancies remain in the magnitude of momentum flux in the spectrum tail and in the sensitivity of QBO characteristics to the source momentum flux (Geller et al. 2013; Alexander et al. 2021).

In reality, convection as a single isolated cell rarely occurs. Convection in tropical and mid-latitude regions often organizes into open- and closed-cellular structures, clusters, or mesoscale convective systems (Jewtoukoff et al. 2013; Preusse et al. 2014). Parameterizations which consider every convective forcing as one equivalent convective source, thereby, miss potential nonlinear interactions among multiple convective cells within a grid box in large-scale models. The idea that such nonlinear interactions appear in the momentum flux expression is not new. Smith and Lin (1982) pointed out nonlinear interactions between mountain and convective forcings, and Seo et al. (2018) revisited the nonlinear interactions. Validation studies show mismatches between parameterized and observed gravity-wave spectra, particularly in the tropics (Alexander et al. 2021), suggesting that convective organization could be a missing ingredient.

This study examines momentum fluxes induced by multiple convective cells. For this, the analytic framework of Seo et al. (2018) is extended to that of multiple convective cells. First, we outline the analytic framework and formulate gravity-wave momentum fluxes induced by multiple convective cells. Then, we analyze the gravity-wave momentum fluxes to show how the multi-cell configuration modifies the momentum flux relative to a

single-cell configuration and we provide its implications for CGWD parameterizations. Finally, the summary and conclusion are given in the last section.

Analytic formulation of multi-cell configuration

In this study, a stably stratified two-layer atmosphere with multiple convective cells is considered (Fig. 1). We consider a two-layer atmosphere in which the lower layer ($0 \leq z \leq H$, where H is the tropopause height) represents the troposphere and has the basic-state wind speed $U(z)$ with constant vertical wind shear ($U(z) = U_0 + sz$, where U_0 is the basic-state wind speed at the surface and s is the basic-state vertical wind shear) and the buoyancy frequency N_T . The upper layer ($z > H$) represents the stratosphere and has the constant basic-state wind speed ($= U_0 + sH$) and the buoyancy frequency $N_S > N_T$. Steady convective forcing is uniformly distributed in the vertical and confined within $h_1 \leq z \leq h_2$, where h_1 and h_2 are the bottom and top heights of convective forcing, respectively. Gravity waves induced by convective forcing propagate upward and transport momentum into the stratosphere. This simplified upper-layer structure is commonly considered in idealized studies to keep the problem tractable and to clarify how the launched momentum flux depends on forcing and basic-state parameters. Further studies including more complex upper-layer structures are needed.

The horizontal distribution of convective forcing for a single convective cell is specified as a heating–cooling pair,

$$q(x) = q_0 \left[\frac{a^2}{x^2 + a^2} - \frac{ab}{x^2 + b^2} \right], \quad (1)$$

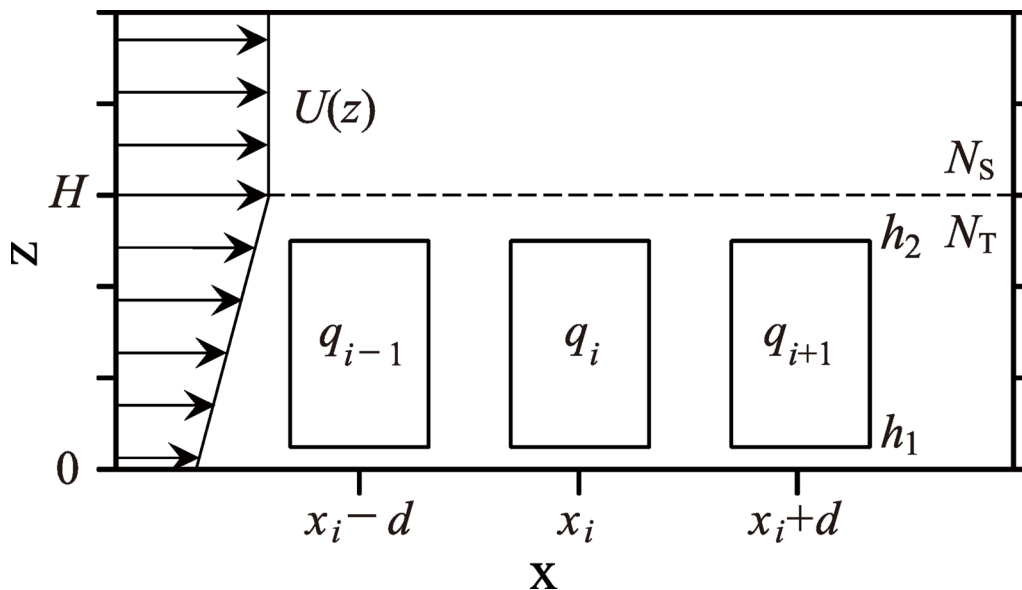


Fig. 1 Structure of a two-layer atmosphere in the presence of multiple convective cells considered in this study

where a is the half-width of the bell-shaped heating function and b is the horizontal width related to the compensating cooling region ($b > a$). This heating-cooling pair naturally produces zero net heating in the horizontal direction. The ratio b/a ($\equiv \beta$) determines how broad the compensating cooling region is relative to the heating region. Seo et al. (2018) used $a = 10$ km and $b = 50$ km ($\beta = 5$) to represent relatively narrow updrafts surrounded by a broad subsiding area. When n identical convective cells exist, the total convective forcing can be written as a sum of individual convective forcings,

$$Q(x) = \sum_{i=1}^n q_i(x - x_{c,i}), \quad (2)$$

where $x_{c,i} = (i-1)d$. Here, d is the spacing between the horizontal centers of convective cells, (see Fig. 1). Small d/a ($\equiv \delta$) indicates closely packed convective cells, whereas large δ represents nearly isolated convective cells. This idealized arrangement could reflect the convective environment in which convective elements are spatially distributed with a characteristic horizontal spacing.

The perturbation fields (u , w , π , b) satisfy the linearized Boussinesq equations under the specified $U(z)$ and $N(z)$.

$$Uu_x + wU_z = -\pi_x, \quad (3)$$

$$\pi_z = b, \quad (4)$$

$$Ub_x + N^2w = \frac{g}{c_p T_0} Q, \quad (5)$$

$$u_x + w_z = 0. \quad (6)$$

Here, u and w are the perturbation horizontal and vertical velocities, respectively, π is the perturbation kinematic pressure, b is the perturbation buoyancy, N is the buoyancy frequency, g is the gravitational acceleration, c_p is the specific heat of air at constant pressure, and T_0 is the reference temperature. For a single convective cell, Seo et al. (2018) solved these equations analytically.

Figure 2 shows the vertical velocity fields obtained by linearly superposing the single-cell solution of Seo et al. (2018) for various combinations of the compensating width β , spacing δ , and number of convective cells n . The vertical velocity fields obtained using the single-cell solution are also shown in Fig. 2. The parameter values used in this study are $H = 12$ km, $h_1 = 1$ km, $h_2 = 11$ km, $U_0 = 10$ m s⁻¹, $s = 10$ m s⁻¹/H, $N_T = 0.01$ s⁻¹, $N_S = 0.02$ s⁻¹, $a = 10$ km, $q_0 = 1$ J kg⁻¹ s⁻¹, and $T_0 = 273.15$ K. The horizontal coordinate is chosen so that for odd n , the cell with the index $i = (n+1)/2$ is located at $x = 0$ and the remaining convective cells are arranged symmetrically around it. Because the governing equations are linear, the vertical velocity field induced by multiple cells is simply the sum of the fields induced by individual convective cells. It is seen from Fig. 2 that the resultant field varies markedly since the interference among adjacent updrafts and downdrafts depends sensitively on β , δ , and n . Detailed

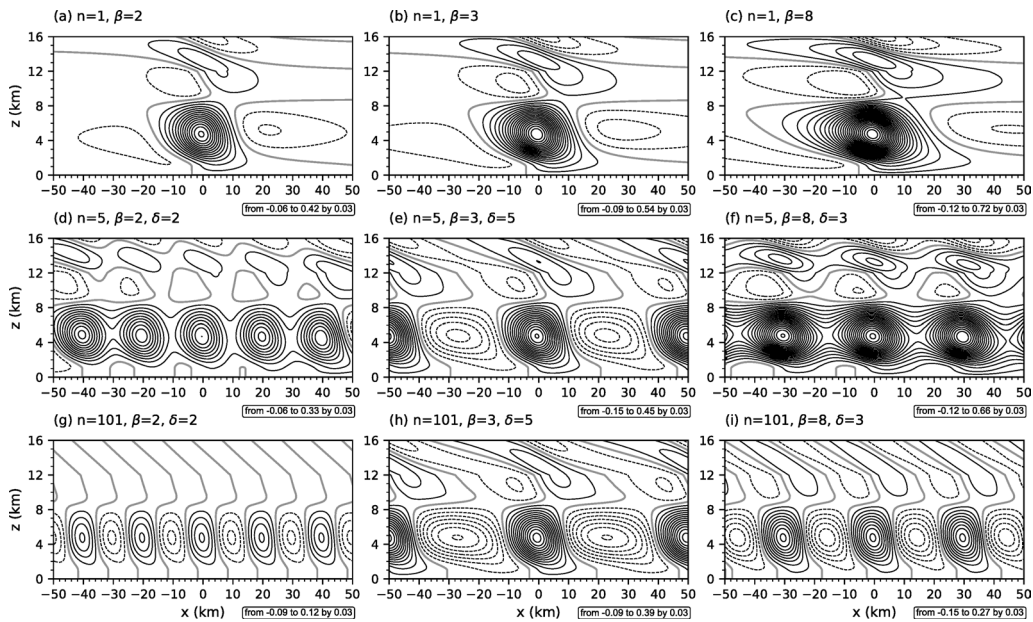


Fig. 2 Convectively forced perturbation vertical velocity fields for **a–c** a single convective cell, **d–f** five convective cells, and **g–i** 101 convective cells. The contour interval is 0.03 m s⁻¹; positive (negative) contours are shown by solid (dashed) lines, and the zero contours are indicated by thick gray lines. The compensating width β , spacing δ , and contour intervals (unit: m s⁻¹) are indicated in each panel

interpretations of the vertical velocity fields are presented in the next section.

The horizontally integrated vertical flux of horizontal momentum M is defined and given as

$$\begin{aligned} M &\equiv \rho_0 \int_{-\infty}^{\infty} u w dx = \rho_0 \int_{-\infty}^{\infty} \left(\sum_i u_i \right) \left(\sum_j w_j \right) dx \\ &= \rho_0 \int_{-\infty}^{\infty} \sum_i (u_i w_i) dx + \rho_0 \int_{-\infty}^{\infty} \sum_{i \neq j} (u_i w_j) dx, \end{aligned} \quad (7)$$

where ρ_0 is the reference density. The first term in the rightmost of Eq. (7) represents the linear superposition of the momentum fluxes for individual convective cells, while the second term arises when multiple convective cells exist and represents the nonlinear mutual interactions among convective cells. Seo et al. (2018) expressed the single-cell momentum flux ($n=1$) as

$$M_S = -\rho_0 U_0^3 N_T^{-1} \mu_{NL}^2 c_1 c_2^2 c_3, \quad (8)$$

where μ_{NL} is the nonlinearity factor for CGWs. The parameters c_1 , c_2 , and c_3 are related to, respectively, the horizontal structure of the convective forcing, vertical structures of the troposphere and convective forcing, and gravity-wave transmission through the tropopause. The parameter $c_1 = \ln[(a+b)^2/(4ab)]$ is determined by the width scales a and b .

For two identical convective cells separated by d , the horizontal interaction term can be evaluated using the horizontal spectral form of the heating-cooling pair $S(k) = e^{-ak} - e^{-bk}$, which leads to an integral of the form $\int_0^\infty (1/k) S^2(k) \cos(kd) dk$. Applying $\int_0^\infty (1/k) e^{-\alpha k} \cos(kd) dk = (1/2) \ln(d^2 + \alpha^2) + \text{const}$, we can obtain the per-cell momentum flux due to mutual interactions between the two convective cells ($M_2/2$) normalized by the single-cell momentum flux (M_S) as follows:

$$\frac{M_2}{2M_S} = 1 + \frac{J(a, b, d)}{J(a, b, 0)}, \quad (9)$$

where $J(a, b, d)$ is the interaction function defined as

$$J(a, b, d) \equiv \frac{1}{2} \left[2 \ln \left(d^2 + (a+b)^2 \right) - \ln \left(d^2 + 4a^2 \right) - \ln \left(d^2 + 4b^2 \right) \right]. \quad (10)$$

It is noted that $J(a, b, 0) = c_1 > 0$. A positive (negative) $J(a, b, d)$ increases (decreases) the magnitude of the momentum flux relative to the single-cell momentum flux. For large d , because $J(a, b, d)$ decays approximately as $1/d^2$, the mutual interactions are weak but occur. For n identical convective cells, it can be shown that the per-cell

momentum flux (M_n/n) normalized by the single-cell momentum flux is expressed as

$$\frac{M_n}{nM_S} = 1 + \frac{2}{n} \sum_{m=1}^{n-1} (n-m) \frac{J(a, b, md)}{J(a, b, 0)} \equiv 1 + J_n(\beta, \delta). \quad (11)$$

Here, $J_n(\beta, \delta)$ represents the nonlinear modification arising from mutual interactions among n distinct convective cells and it is called the nonlinear modification factor hereafter. For $n=1$, that is, the single-cell configuration, $M_1 = M_S$ since $J_1 = 0$ (Eqs. (8) and (11)). For $n=2$, that is, the two-cell configuration, Eq. (11) becomes Eq. (9). For $n > 1$, the sign and magnitude of J_n determine whether the momentum flux is enhanced or suppressed relative to the linear superposition of the momentum fluxes induced by individual convective cells.

Results and discussion

Properties of the nonlinear modification factor

Figure 3 shows how the nonlinear modification factor $J_n(\beta, \delta)$ varies with the compensating width β and spacing δ for different numbers of convective cells n . Here, n is chosen as 2, 5, 101, and ∞ . Although $n=101$ or ∞ are not realistic values, they are considered to examine the mathematical behaviors of J_n . Positive (negative) values of J_n indicate enhancement (suppression) of momentum flux compared with the single-cell momentum flux.

In the two-cell configuration ($n=2$, Fig. 3a), the nonlinear modification factor is given by $J_2(\beta, \delta)$. The sign of $J_2(\beta, \delta)$ changes across the zero contour where $J_2(\beta, \delta) = 0$. The transition curve (i.e., zero contour) $\delta_0(\beta)$ is analytically given by

$$\delta_0(\beta) = \sqrt{\frac{(1+\beta)^4 - 16\beta^2}{2(1-\beta)^2}}. \quad (12)$$

Equation (12) represents the condition at which momentum-flux enhancement and suppression effects balance each other (black-dashed line in Fig. 4). $\delta_0(\beta)$ increases monotonically with β . For a given β , J_2 is positive for $\delta < \delta_0$ and is negative for $\delta > \delta_0$. As β increases and δ decreases, J_2 increases, that is, mutual interactions between two convective cells become strong. In the large separation limit ($\delta \rightarrow \infty$), $J_2 \rightarrow 0$ and $M_2/(2M_S) \rightarrow 1$.

When the number of cells increases ($n > 2$, Fig. 3b–d), each convective cell mutually interacts with other convective cells with separations md weighted by $(n-m)$. For $n=5$, a narrow region of strong positive nonlinear modification is seen in the parameter space of large β and small δ . This indicates that as n increases, the conditions for a positive mutual-interaction effect become more restrictive, favoring a broader compensating region and more closely packed cells. For $n=101$, negative nonlinear

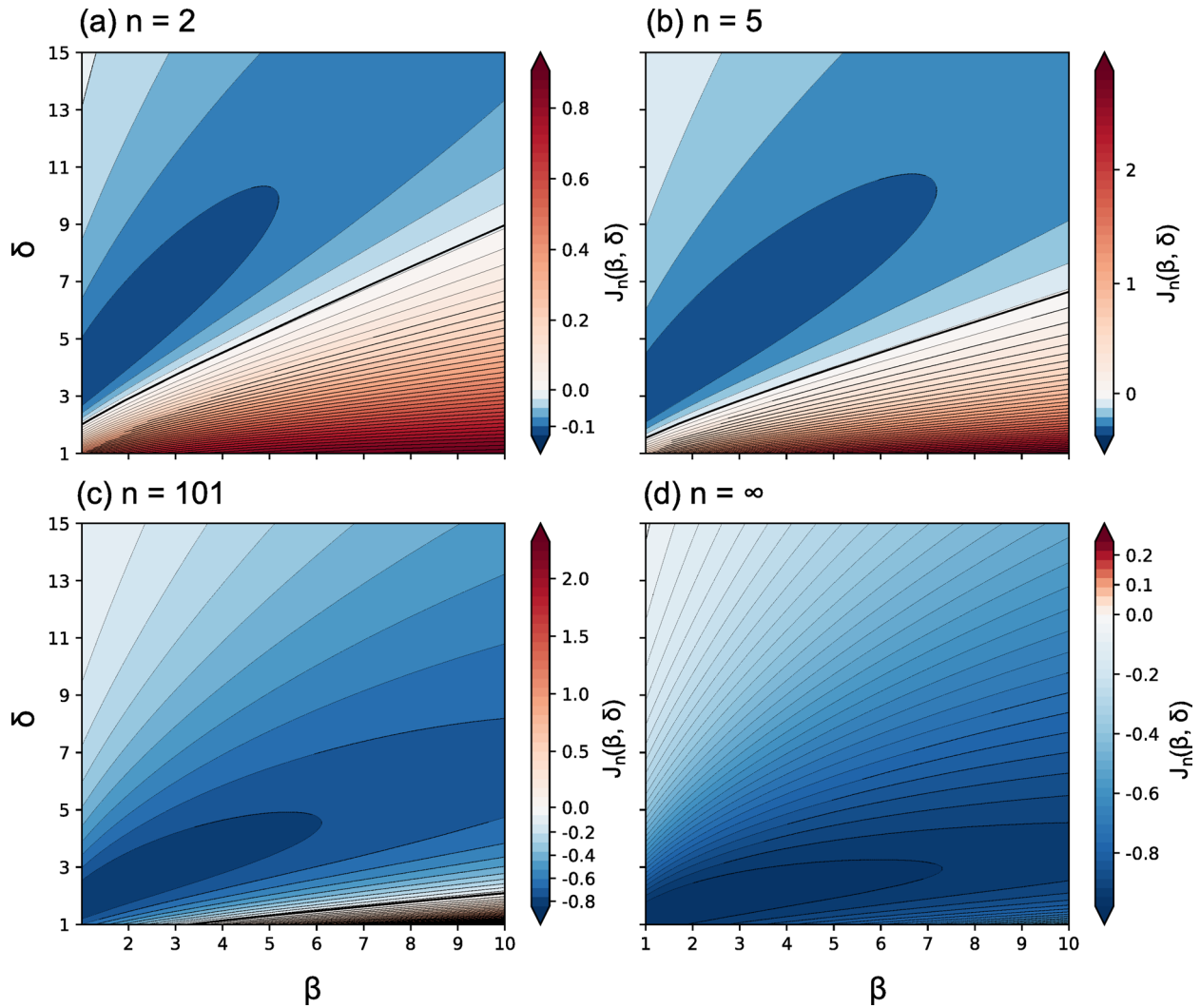


Fig. 3 Nonlinear modification factor $J_n(\beta, \delta)$ for **a** $n=2$, **b** $n=5$, **c** $n=101$, and **d** $n=\infty$

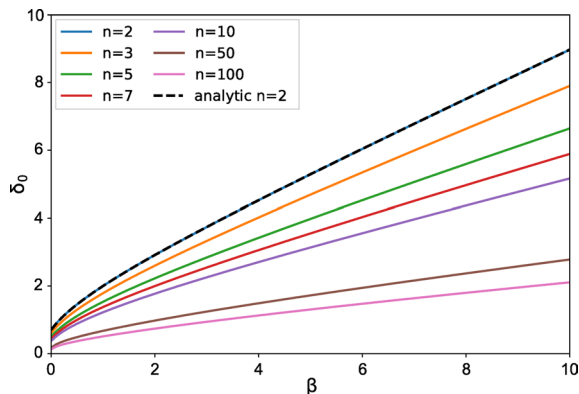


Fig. 4 Numerically obtained transition curve $\delta_0(\beta)$ separating the positive and negative regions of $J_n(\beta, \delta)$ for different values of n . The analytically obtained δ_0 for $n=2$ from Eq. (12) is plotted as the black dashed line

modification appears across most of the parameter space of β and δ . In the asymptotic limit of $n \rightarrow \infty$, the nonlinear modification is negative in the entire parameter space of β and δ (Fig. 3d). In short, Fig. 3 and Eq. (11) show that closely packed convective cells with broad subsidence act to enhance the per-cell momentum flux, while widely spaced convective cells act to reduce the per-cell momentum flux.

The change of the transition curve $\delta_0(\beta)$ with the number of convective cells is presented in Fig. 4. Here, large values of n are included to examine the mathematical properties of $\delta_0(\beta)$. As the number of convective cells increases, the transition curve shifts toward smaller δ . This means that the region of positive J_n shrinks and the region of negative J_n expands.

Flow structures and mutual interactions in multi-cell configuration

The vertical velocity fields (Fig. 2) provide some insights into how the nonlinear modification factor influences the momentum flux through mutual interactions among convective cells. In the single-cell configuration ($n=1$, Fig. 2a–c), the vertical velocity field is simply that of Seo et al. (2018) and reflects the imposed heating–cooling pair. For a stationary forcing, upward propagating gravity waves for $U>0$ exhibit an upwind tilt of the phase lines, implying an upward wave energy propagation (e.g., Nappo 2002; Fritts and Alexander 2003). As β increases from 2 to 8 (Fig. 2a–c), the total cooling is kept the same but distributed over a broader region. With this broader cooling distribution, the gravity-wave response above the cloud-top height becomes stronger and the updrafts in the heating core become stronger and are more horizontally extended. The maximum vertical velocity increases with increasing β . As β increases, the vertical velocity field is more asymmetric in the direction of the basic-state wind in the convective forcing region and the upwind-tilted updraft region is more pronounced above the cloud-top height.

In the five-cell configuration, ($n=5$, Fig. 2d–f), the individual convective cells interact with one another. For the more compact cases (Fig. 2d, f), updraft cores and subsidence regions strongly overlap, producing a merged updraft region and no subsidence between neighboring convective cells. In contrast, the intermediate case (Fig. 2e) shows clearly separated updraft–downdraft patterns along the array of convective cells. These differences directly demonstrate how strongly adjacent convective cells can interact. The compact cases of small δ correspond to the positive- J_n region, whereas the more widely spaced case of large δ lies in the negative- J_n region (Fig. 3b).

In the 101-cell configuration ($n=101$, Fig. 2g–i), all three combinations of (β, δ) produce nearly periodic, well-separated updraft–downdraft patterns. In this configuration, increasing β no longer monotonically increases the maximum vertical velocity. The intermediate case (Fig. 2h) exhibits stronger updraft than the other cases (Fig. 2g, i). These results are consistent with that increasing n causes the positive- J_n region for small δ to shrink and results in a negative nonlinear modification over most of the parameter space of β and δ . In other words, the merged updraft region without subsidence (Fig. 2d, f) is changed to the alternating updraft–downdraft patterns (Fig. 2g, i), resulting in a suppression of the momentum flux. While the overall stratospheric wave pattern in the multi-cell cases still exhibits an upwind phase tilt for $U>0$ in Fig. 2, this structural signature alone is not sufficient to determine whether the net momentum flux is enhanced or suppressed. This motivates a direct

diagnosis of the momentum flux using the running-sum decomposition introduced below.

To examine how the flow structure affects the momentum flux, the running sum of momentum flux is defined and calculated as

$$\begin{aligned} R(x, z) &\equiv \rho_0 \int_{-\infty}^x u(\xi, z) w(\xi, z) d\xi \\ &= \rho_0 \int_{-\infty}^x \sum_i u_i w_i d\xi + \rho_0 \int_{-\infty}^x \sum_{i \neq j} u_i w_j d\xi \quad (13) \\ &= R_S + R_C. \end{aligned}$$

Figure 5 presents the calculated running sum of the per-cell momentum flux for the nine cases corresponding to Fig. 2. The top row shows the single-cell R_S/n , and the middle and bottom rows show the mutual interaction term R_C/n decomposed from the total running sum R/n . The running sum is normalized by the maximum value of $|R_S/n|$ or $|R_C/n|$ divided by the reference density ρ_0 . At each height, the sign of R (and its components R_S and R_C) indicates the sign of the horizontally accumulated (from $-\infty$ to x) vertical flux of horizontal momentum. For combinations of (β, δ) in the negative- J_n region of Fig. 3 (Fig. 5e, h), the signs of R_C/n in the upper-troposphere to lower-stratosphere are the opposite in the vertical to those of R_S/n for the single convective cell (Fig. 5b) downstream of the rightmost convective cell and result in a suppression of R/n there. For combinations of (β, δ) in the positive- J_n region of Fig. 3 and for $n=5$ (Fig. 5d, f), R_C/n shares the same signs in the vertical as those of R_S/n for the single convective cell (Fig. 5a, c) downstream of the rightmost convective cell and yield an enhancement of R/n there. When n changes from 5 to 101, the vertical wavy pattern of R_C/n downstream of the rightmost convective cell changes phase from being mostly in phase (Fig. 5d, f) with the vertical wavy pattern of R_S/n (Fig. 5a, c) to being out of phase (Fig. 5g, i). This phase change turns the enhancement of the momentum flux for $n=5$ into the suppression of the momentum flux for $n=101$, consistent with the change from the positive- J_n for $n=5$ to negative- J_n for $n=101$ in Fig. 3. Taken together, Figs. 2, 3, 4 and 5 indicate that purely geometric differences in convective organization, as measured by the parameters β , δ , and n , systematically tangle the flow structure and reshape R_C , hence ultimately strengthening or weakening the gravity-wave momentum flux.

Implications for CGWD parameterizations

The analytic framework developed in this study has some implications for the parameterization of CGWD. The existing CGWD parameterizations (e.g., Chun and Baik 1998) calculate the cloud-top momentum flux using a single convective source in a grid box ($J_1=0$). In

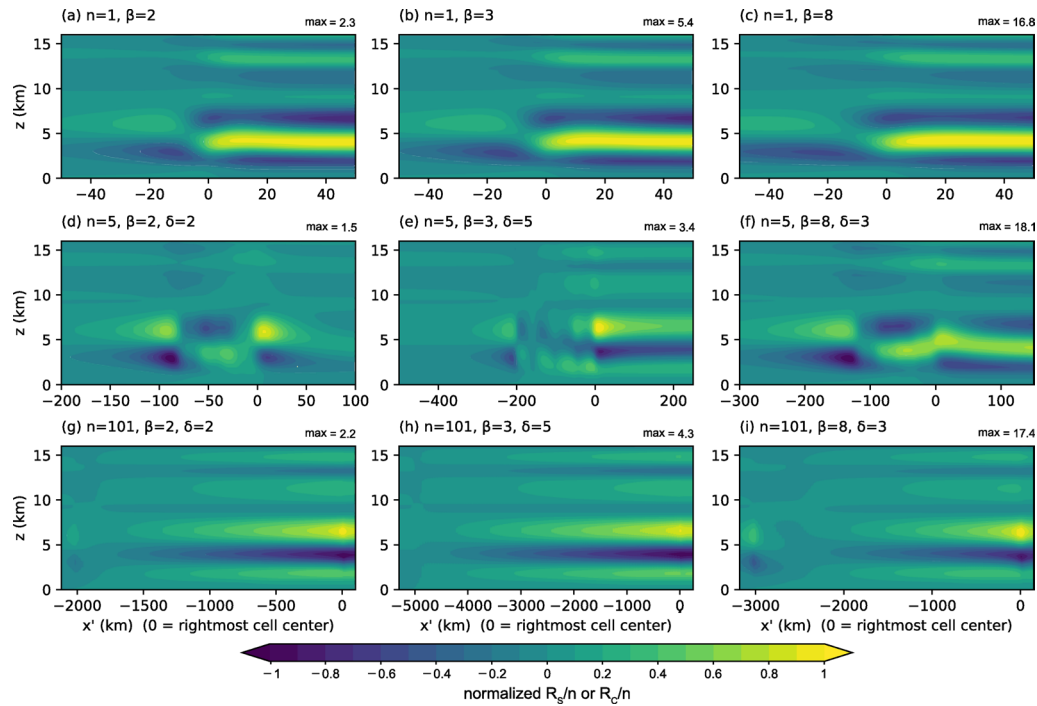


Fig. 5 Per-cell running sum of the vertical flux of horizontal momentum for **a–c** a single convective cell, **d–f** five convective cells, and **g–i** 101 convective cells normalized by the maximum values of $|R_g/n|$ or $|R_c/n|$ divided by the reference density ρ_0 (unit: $\text{m}^3 \text{s}^{-2}$). In the horizontal coordinate of each panel, $x'=0$ indicates the center of the rightmost convective cell. The compensating width β , spacing δ , and the maximum values of $|R_g/n|$ or $|R_c/n|$ divided by the reference density ρ_0 are indicated in each panel

this study, it is shown that the per-cell momentum flux induced by multiple convective cells (multiple sources in a grid box) is equal to the single-cell momentum flux multiplied by $[1 + J_n(\beta, \delta)]$, where $J_n(\beta, \delta)$ is the nonlinear modification factor arising from mutual interactions among distinct convective cells. In principle, it would be straightforward to include the effect of convective organization in the cloud-top momentum flux: multiply the single-cell momentum flux by $[1 + J_n(\beta, \delta)]$ without changing the rest of the parameterization.

For the geometric properties of convective organization (β , δ , and n), a practical way would treat these parameters as free parameters characterizing the organization of subgrid-scale convection. Some properties of convective organization could be deduced by diagnostics of the results from a sub-grid scale convection scheme. For randomly distributed convective cells, deducing β , δ , and n would require additional assumptions and validations, which is beyond the scope of the present idealized study. Other properties of convective organization could not be explicitly deduced. Even when relevant properties are available from model diagnostics, relating them quantitatively to β , δ , and n is very difficult, needing further in-depth investigation. At present, parameter tuning or parameter optimization might be a way of determining the free parameters.

Once the parameters β , δ , and n are determined, Eq. (11) provides $J_n(\beta, \delta)$ as a purely geometric modification factor. This enables the CGWD to respond not only to the intensity of convection but also to how convective cells are organized horizontally. The analytic framework in this study shows that convective organization affects the momentum flux obtained using a CGWD parameterization and that this influence is represented by a simple multiplicative modification term.

Summary and conclusion

This study examines how the horizontal organization of convection modifies gravity-wave momentum flux when multiple convective cells exist. Within the two-layer analytic framework of Seo et al. (2018), the vertical flux of horizontal momentum per cell induced by identical convective cells is expressed as that induced by a single convective cell times $[1 + J_n(\beta, \delta)]$. Here, $J_n(\beta, \delta)$ is the nonlinear modification factor arising from mutual interactions among distinct convective cells, β is the ratio of compensating-cooling width to heating-core width, δ is the nondimensional spacing between neighboring cells, and n is the number of convective cells. J_n vanishes for a single cell.

The analysis shows that compactly packed cells with broad cooling (large β and small δ) act to enhance the per-cell momentum flux compared with the single-cell

momentum flux. This is because the heating and cooling patterns of neighboring cells strongly overlap and the heating–cooling pairs effectively behave like a single, broader forcing region that emits stronger gravity waves. As the spacing between the neighboring convective cells increases, updraft–downdraft couplets become more distinct. As the number of convective cells increases, the positive- J_n region shrinks toward smaller δ and disappears in the limit of infinite number of convective cells.

The findings of this study suggest one possible way to incorporate subgrid-scale convective organization into CGWD parameterizations. Specifically, $[1 + J_n(\beta, \delta)]$ is used as a multiplicative modification term to the single-cell momentum flux which is calculated using host CGWD parameterization and sub-grid scale convective scheme, providing a pathway for CGWD to respond to changes in the geometric properties of convective organization. The implementation and verification of this modification in a general circulation model deserve future investigation. To get further insights into the geometric properties of convective organization, cloud-resolving simulations under various environmental settings and observations in the context of the analytic modification presented in this study are needed, which will help to develop CGWD parameterizations further. The present framework provides an extension of single-cell idealized studies to a multi-cell configuration. Configurations with more complex convective organizations, considering heterogeneous cell structures and systematically varying cell depth or cloud-top height, deserve future investigation.

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Author contributions

J. M. Seo and J.-J. Baik both designed this study. J. M. Seo developed the analytic formulation and wrote the original manuscript. All authors reviewed and revised the manuscript.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare that they have no competing interests.

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